Relational Algebra

txt2tex example

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$[Title, Author, MemberName, Genre]$

$$\begin{array}{|l}
\multicolumn{1}{l}{\textit{Book}} \\ \hline
bookId : \mathbb{N} \\
title : Title \\
author : Author \\
genre : Genre \\
pages : \mathbb{N} \\
year : \mathbb{N} \\ \hline
\end{array}$$

$$\begin{array}{|l}
\multicolumn{1}{l}{\textit{Member}} \\ \hline
memberId : \mathbb{N} \\
name : MemberName \\ \hline
\end{array}$$

$$\begin{array}{|l}
\multicolumn{1}{l}{\textit{Loan}} \\ \hline
loanId : \mathbb{N} \\
bookId : \mathbb{N} \\
memberId : \mathbb{N} \\
dueYear : \mathbb{N} \\ \hline
\end{array}$$

Restriction (sigma)

$\mathrm{Restrict}_{pages \geq 500}(Book)$
$\mathrm{Restrict}_{year \geq 1990}(\mathrm{Restrict}_{pages \geq 500}(Book))$

Projection (pi)

$\mathrm{Project}\{title, author\}(Book)$
$\mathrm{Project}\{title, author\}(\mathrm{Restrict}_{pages \geq 500}(Book))$

Renaming (rho)

$\mathrm{Rename}_{bookId \to id}(Book)$
$\mathrm{Rename}_{bookId \to id, author \to writer}(Book)$

Natural Join and Theta - Join (bowtie)

$Loan \otimes Book$
$\mathrm{Rename}_{bookId \to id}(Loan) \otimes \mathrm{Rename}_{bookId \to id}(Book)$
$\mathrm{Join}_{Member.memberId = Loan.memberId}(Member, Loan)$

Division (div)

$Loan$ div $Book$

Combined Query

$\text{Project}\{title, author\}(\text{Restrict}_{pages \geq 500}(Book))$